

Practical Experience with the Use of Shiftwave Technology in Working with Children with Autism Spectrum Disorders: Clinical Observations and Possible Neurophysiological Mechanisms

Introduction

This report summarizes the practical experience of using the Shiftwave chair at the “Rivnova” Psychological Assistance Center (Kyiv, Ukraine) in working with children and adolescents, primarily those with autism spectrum disorders (ASD). The document is based on clinical observations by the Center’s practicing psychologists and aims to describe the specifics of applying Shiftwave technology as part of comprehensive psychological, neuropsychological, and sensory-integrative support.

In modern practice for supporting children with ASD, attention is focused not only on the development of cognitive, speech, and behavioral skills, but also on creating a functional state conducive to learning, therapy, and social interaction. A child’s readiness for psychocorrectional work depends to a large extent on their ability to regulate their arousal level, sustain attention, process sensory information, and recover from stress.

This is particularly important right now in Kyiv, where children often arrive after sleepless nights in bomb shelters.

Shiftwave combines rhythmic vibrational stimulation, audio accompaniment, and somatosensory stimulation of the body. This combination can provide additional sensory input and create conditions for regulating physiological arousal levels. At the same time, the mechanisms by which the technology exerts its effects have not been directly studied within the Center’s work; therefore, the neurophysiological explanations provided below are based on well-founded hypotheses.

At the “Rivnovaga” Center, Shiftwave is used as a supplementary component of comprehensive programs for children with ASD, attention-deficit/hyperactivity disorder (ADHD), sensory processing differences, high levels of anxiety, and difficulties with emotional self-regulation. The technology is not considered a standalone therapeutic method and is combined with psychotherapy, neuropsychological intervention, sensory integration, play therapy, bioacoustic intervention, and other methods selected according to the child’s individual needs.

This report combines practical observations by the Center’s specialists with current understanding of proprioception, sensory integration, autonomic regulation, and interoception. The presented results do not constitute proof of Shiftwave’s clinical efficacy; however, they may serve as a foundation for further systematic research.

Organization of Shiftwave’s Practical Use

In the Center’s practice, a Shiftwave session is conducted before or after the main session, depending on the child’s current condition and individual needs.

Before therapy, the chair is used to facilitate the transition from an educational or everyday environment to structured work. After an intensive psychological, neurocorrectional, or sensory-integration session, it is used as an adjunct to aid recovery and reduce residual tension.

The most commonly used program is a 10-minute session in Ukrainian, adapted for children. In the Center's practice, this duration is generally well tolerated; however, the parameters of the procedure are determined on an individual basis, taking into account the child's age, sensory profile, emotional state, and reactions.

Different components of the program may be tolerated to varying degrees. Some children respond positively to the combination of vibration and audio accompaniment, while others refuse to wear headphones but respond well to vibrational stimulation. This underscores the need for separate adjustment of the auditory and vibrational components.

Specifics of Use in Working with Children and Adolescents with ASD

The Center has accumulated the most practical experience in working with children and adolescents with ASD. According to psychologists' observations, most of them adapt quickly to the procedure, rarely show significant resistance, and often express a desire to repeat the session during subsequent visits.

The technology is particularly valuable in working with adolescents. At younger ages, additional proprioceptive and somatosensory input can be provided through active games, resistance exercises, deep pressure, or other forms of sensory interaction. When working with adolescents, the use of such methods is sometimes less convenient or acceptable. Shiftwave allows for rhythmic mechanical stimulation over a large area of the body without the need for direct physical contact with the therapist.

At the same time, the responses of children with ASD vary. Some actively seek deep bodily sensations, while others may be sensitive to vibration, sound, or intense somatosensory stimulation. Therefore, the use of this technology requires a gradual introduction, monitoring of nonverbal reactions, and individualized selection of parameters.

Use Before Therapy Sessions

A significant number of children visit the Center after the school day, while in a state of physical, cognitive, or sensory exhaustion. For a child with ASD, being at school may involve having to adapt to noise, social interaction, frequent switching between activities, and a large number of external stimuli. Or after nighttime shelling and sleepless nights.

In such cases, a short Shiftwave session is used before the start of a psychological or neurocorrectional session. According to specialists' observations, after the

session, some children exhibited:

- a reduction in motor and emotional tension;
- a faster transition to interaction with the psychologist;
- longer sustained attention;
- better comprehension of instructions;
- easier engagement in structured activities.

These changes were not measured using standardized scales or physiological indicators; therefore, they are considered clinical observations rather than a proven therapeutic effect.

Use Following Intensive Sessions

After neuropsychological, sensory-integration, or other intensive sessions, some children showed signs of overexertion, motor agitation, or, conversely, pronounced exhaustion.

According to psychologists' observations, the use of Shiftwave after a session was accompanied in some children by a gradual reduction in tension and a calmer conclusion to the visit. Parents and specialists also noted that after the procedure, children sometimes coped more easily with subsequent daily demands, such as the trip home, homework, or additional activities.

At the same time, these observations require systematic verification, as many factors can influence a child's condition after therapy: the intensity of the session, the level of fatigue, the time of day, the individual sensory profile, and the overall emotional state.

Regulation of Activation Level

In the Center's practice, vibration intensity is selected according to the child's current state. Lower intensity levels are more often used in cases of increased tension or overload, while higher levels may be applied when it is necessary to increase the level of arousal before cognitive or motor tasks.

According to specialists' observations, intensities up to 50% were more frequently associated with:

- a reduction in psycho-emotional stress;
- behavioral stabilization;
- a reduction in motor arousal;
- preparation for calm therapeutic activities.

In some cases, an intensity of over 70% was used to increase alertness, work capacity, and readiness to perform tasks.

These limits should not be considered universal clinical recommendations. The response to intensity is individual, so the parameters of the procedure should be selected gradually, taking into account the child's behavioral and physical responses.

Observation of Somatosensory and Motor Responses

While the chair was in operation, psychologists often observed a gradual decrease in chaotic or excessive movements, stabilization of posture, and the child's focus on bodily sensations. After the session ended, some children transitioned more easily to organized activities and demonstrated a reduced need for constant motor exploration.

Such manifestations may be associated with additional proprioceptive and somatosensory input. At the same time, it is impossible to assert that "satisfaction of proprioceptive needs" occurs in all cases, since the sensory profile of children with ASD is individual, and reactions to vibration can vary significantly.

A practically important observation was the possibility of using the vibration component independently. Some children did not respond to the audio accompaniment but maintained a positive or neutral reaction to the mechanical stimulation. This indicates the value of flexibly tailoring the procedure.

Use as Part of Comprehensive Care

At the Shiftwave Center, it is used in conjunction with psychological, neuropsychological, and sensory-integration methods. Specifically, the chair is combined with:

- neuropsychological therapy;
- sensory integration work;
- play therapy;
- bioacoustic therapy;
- exercises based on the Belgaou method.

The Center's specialists subjectively assess some of these combinations as helpful in preparing a child for a class or wrapping up intensive work. In particular, after the sequential application of bioacoustic correction and Shiftwave, some children exhibited marked physical and emotional relaxation.

However, the additional effect of each individual component was not investigated. Therefore, the positive changes observed within the comprehensive program cannot be attributed exclusively to the use of Shiftwave.

Clinical Observations

Clinical Observation 1. A teenager with ASD after the school day

A teenager with ASD regularly attended sessions with a psychologist after the

school day ended. Before the sessions, he often showed signs of exhaustion, had difficulty concentrating, and required a long time to engage in interaction.

A 10-minute Shiftwave session was used before the main session. After the session, the psychologist observed a reduction in motor tension, faster establishment of contact, better comprehension of instructions, and longer attention span while performing tasks.

The practical value of the technology in this case also lay in its ability to provide additional somatosensory and proprioceptive stimulation without direct physical contact.

This observation does not allow us to establish a cause-and-effect relationship between the procedure and changes in the adolescent's behavior, but it illustrates one possible use of Shiftwave prior to therapy.

Clinical Observation 2. A Child with Difficulties Controlling Physiological Needs

Of particular interest is the case of a four-year-old child with significant difficulties in independent defecation. At the start of treatment, the child was practically unable to defecate without assistive procedures.

The comprehensive program included play therapy, sensory integration work, bioacoustic correction, and regular use of Shiftwave.

During the course of treatment, a gradual improvement in the child's independent control of physiological needs was observed. By the final stage, the child had almost completely transitioned to using the toilet independently.

Since the intervention was comprehensive, it is impossible to determine Shiftwave's specific contribution to the observed progress. This case does not confirm the technology's effect on interoception or autonomic regulation, but it may be relevant for formulating hypotheses and conducting further research.

Summary of Clinical Observations

According to observations by the Center's specialists, most children with ASD who were offered Shiftwave responded well to the procedure and adapted to it quickly. Most often, following sessions, there was a reduction in motor and emotional tension, more organized behavior, easier transitions between activities, and greater readiness to interact with the psychologist.

The nature of the response depended on the child's initial state, the intensity of the vibration, tolerance of the audio accompaniment, and the timing of the procedure within the session. In the Center's practice, Shiftwave was used before therapy to facilitate engagement in the session or after an intense workload—to support recovery.

The results presented here are clinical observations. They do not allow for establishing a cause-and-effect relationship between the use of Shiftwave and the observed changes. At the same time, the repeatability of certain reactions justifies

the need for further systematic study of the technology.

Clinical Observations and Possible Mechanisms

Specialists' Observations	Possible explanation
Reduction in motor and emotional tension	Regulation of physiological arousal levels and additional somatosensory input
More stable posture and reduced motor search	Proprioceptive and tactile stimulation
Easier integration into structured activities	Reduced overload and optimized arousal levels
Longer attention span	A more favorable functional state prior to task performance
Greater readiness for interaction	Reduced tension and improved behavioral organization

The explanations provided are hypothetical. They have not been confirmed by direct neurophysiological measurements during the use of Shiftwave.

Possible neurophysiological mechanisms

Sensory processing in ASD

ASD is associated not only with characteristics of social communication and behavior but also with differences in the perception and processing of sensory information.

Children with ASD may exhibit hyper-reactivity, hypo-reactivity, or an active seeking of certain sensory stimuli. These characteristics may involve the auditory, tactile, vestibular, visual, and proprioceptive systems.

Sensory overload or poor regulation of sensory input can make it difficult to concentrate, switch between activities, learn, and interact socially. Therefore, one of the goals of sensory-oriented support is not simply to “calm” the child, but to help establish a functional state in which the child can participate more effectively in therapeutic or educational activities.

Below are several possible mechanisms that may be relevant to the changes in arousal level, motor organization, attention, and readiness for interaction observed by the Center’s specialists. None of these mechanisms has yet been confirmed by direct research conducted by Shiftwave.

Somatosensory and Proprioceptive Input

One possible mechanism of Shiftwave’s effect is additional somatosensory and proprioceptive input resulting from rhythmic mechanical stimulation of the body.

The proprioceptive system receives information from muscles, tendons, and joints and is involved in body schema formation, postural control, movement coordination, and motor planning. The somatosensory system also processes touch,

pressure, vibration, and other bodily signals.

Shiftwave's rhythmic vibration can activate skin, muscle, and other mechanoreceptors across a large area of the body. For some children with ASD who actively seek deep bodily sensations, this stimulation can be organizing or calming.

At the same time, the response to proprioceptive and vibratory stimulation varies from child to child. Not all children with ASD require intense sensory input, and for some, it may be unpleasant or overwhelming.

Sensory Integration

Sensory integration refers to the nervous system's ability to receive, distinguish, integrate, and use information received through various sensory channels.

Shiftwave combines vibratory, somatosensory, and auditory stimulation. In theory, a controlled and predictable combination of these stimuli can help some children better organize their sensory experiences and transition to subsequent activities.

However, multimodal stimulation is not always beneficial on its own. For children with high sensory sensitivity, the simultaneous exposure to multiple stimuli can increase their sensory overload. That is why it is essential to individually select parameters, introduce the procedure gradually, and allow for the option to turn off individual components.

Regulation of the Level of Physiological Arousal

Effective participation in learning and therapy requires an optimal level of arousal. Excessive arousal can manifest as anxiety, motor agitation, irritability, or avoidance. Insufficient arousal may be accompanied by lethargy, slow engagement in activities, and difficulty sustaining attention.

Rhythmic and predictable sensory stimulation can influence subjective levels of tension and alertness. In the Center's practice, lower intensity was more often used to promote calmness, while higher intensity was used to promote arousal.

It can be assumed that the Shiftwave effect depends in part not on the specific action of a single brain structure, but on a general change in the level of arousal, which makes the child's state more conducive to further activity.

This hypothesis requires verification using standardized behavioral scales and physiological indicators.

Autonomic Nervous System

Some children with ASD experience difficulties with autonomic regulation, which may manifest as increased physiological reactivity, hypervigilance, or slow recovery after exertion.

Rhythmic vibration and soothing background music may potentially help reduce subjective tension and facilitate a transition to a more stable physiological state.

However, there is currently no evidence to suggest that the Shiftwave directly activates the parasympathetic nervous system or improves vagal regulation.

Within the framework of the polyvagal model, a reduction in stress and an increase in readiness for interaction can be interpreted as a transition to a state of greater physiological safety. At the same time, this model is theoretical, and such an interpretation does not confirm a direct effect of Shiftwave on vagal function.

To assess the potential impact on autonomic regulation, it is advisable to use objective measures, including heart rate, heart rate variability, respiratory rate, and electrodermal activity.

Interoception and Body Awareness

Interoception is the perception and interpretation of signals originating from the body's internal environment. It is associated with recognizing hunger, thirst, pain, heartbeat, the need to use the restroom, and the bodily components of emotions.

Some children with ASD may have difficulty recognizing or interpreting such signals. Additional somatosensory input has the potential to increase attention to bodily sensations; however, there is insufficient evidence to conclude that vibratory stimulation directly improves interoceptive accuracy.

The case described in the practical section involving a child with difficulties controlling physiological needs may be relevant for further study of interoceptive processes. At the same time, it does not allow for a conclusion regarding the causal effect of Shiftwave, since the child was receiving several types of support simultaneously.

The Possible Role of the Cerebellum

The cerebellum is involved not only in movement coordination but also in motor learning, predicting the sensory consequences of movement, regulating attention, and interacting with other brain networks.

Since the cerebellum receives a significant amount of proprioceptive information, it can be assumed that rhythmic somatosensory stimulation indirectly engages systems associated with postural control, motor organization, and sensorimotor coordination.

However, clinical behavioral observations do not allow us to determine whether the observed changes are specifically related to cerebellar function. Therefore, claims regarding “ ” “cerebellar stimulation,” or enhanced interaction between the cerebellum and the cortex require direct experimental confirmation.

At the Shiftwave Center, this approach is also combined with exercises based on the Belgau method. Specialists subjectively assess this combination as beneficial for some children; however, its additional effect has not been studied separately.

Interhemispheric and Network Interaction

RAS is associated with complex and heterogeneous differences in functional

interaction between various brain networks. These differences cannot be reduced solely to insufficient interaction between the right and left hemispheres.

In theory, multimodal sensory stimulation engages several information-processing systems simultaneously. However, there is currently no data to confirm that Shiftwave improves interhemispheric interaction, corpus callosum function, or neural network connectivity.

Therefore, this mechanism should be considered only as a research hypothesis that requires neurophysiological or neuroimaging verification.

Shiftwave as a Means of Preparing for Therapy

Based on the practical experience of the Shiftwave Center, it is more appropriate to view Shiftwave not as a standalone treatment for ASD, but as an adjunctive tool for regulating the patient's state prior to therapy or for recovery after exertion.

Its potential value may lie in creating a predictable sensory environment, providing additional somatosensory input, and allowing for individualized regulation of the level of arousal.

In favorable cases, this may facilitate:

- the transition from overstimulation to structured activities;
- maintaining attention;
- understanding instructions;
- behavioral organization;
- readiness to interact with a specialist;
- completing an intensive session.

These potential effects should be assessed individually for each child, without generalizing them to all children with ASD.

Limitations of Clinical Observations

The presented material has a number of limitations.

First, the observations were conducted in a routine clinical setting without a control group, randomization, or blinding.

Second, changes were assessed primarily by psychologists and parents without the use of standardized scales, instrumental measurements, or independent evaluation.

Third, Shiftwave was used as part of comprehensive programs, so it is impossible to isolate its effect from the effects of psychotherapy, neuropsychological intervention, sensory integration therapy, bioacoustic intervention, and other components.

Fourth, the children's response may have depended on nonspecific factors: the

novelty of the procedure, the specialist's expectations, the nature of the interaction, the time of day, the level of fatigue, and natural variability in their condition.

Fifth, in clinical practice, there has been no systematic comparison of different regimens regarding intensity, duration, frequency of use, and sequence of procedures.

Therefore, the presented results should be interpreted as preliminary clinical observations that help formulate research questions but do not confirm the effectiveness or specific mechanisms of action of the technology.

Conclusions

The practical experience of the “Rivnovaga” Psychological Assistance Center indicates that Shiftwave may be suitable for some children and adolescents with ASD as an adjunct component of comprehensive care.

In the Center's practice, the technology was used before therapy sessions to facilitate engagement and after intensive sessions to support recovery. Most often, specialists observed a reduction in motor and emotional tension, more organized behavior, easier transitions between activities, and greater readiness for interaction.

The most likely hypotheses regarding the mechanisms of action are related to additional somatosensory and proprioceptive input, regulation of physiological arousal levels, and the creation of a predictable sensory environment.

At the same time, the available clinical observations do not allow for establishing a causal relationship between the use of Shiftwave and the observed changes. Nor has a direct effect of the technology on autonomic regulation, interoception, cerebellar function, interhemispheric interaction, or neural connectivity been confirmed.

Further research on Shiftwave should be conducted using standardized behavioral measures, physiological measurements, control conditions, and a clear description of the procedure's parameters. It is particularly important to determine for which sensory profiles, age groups, and functional states the technology may be most suitable and beneficial.

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